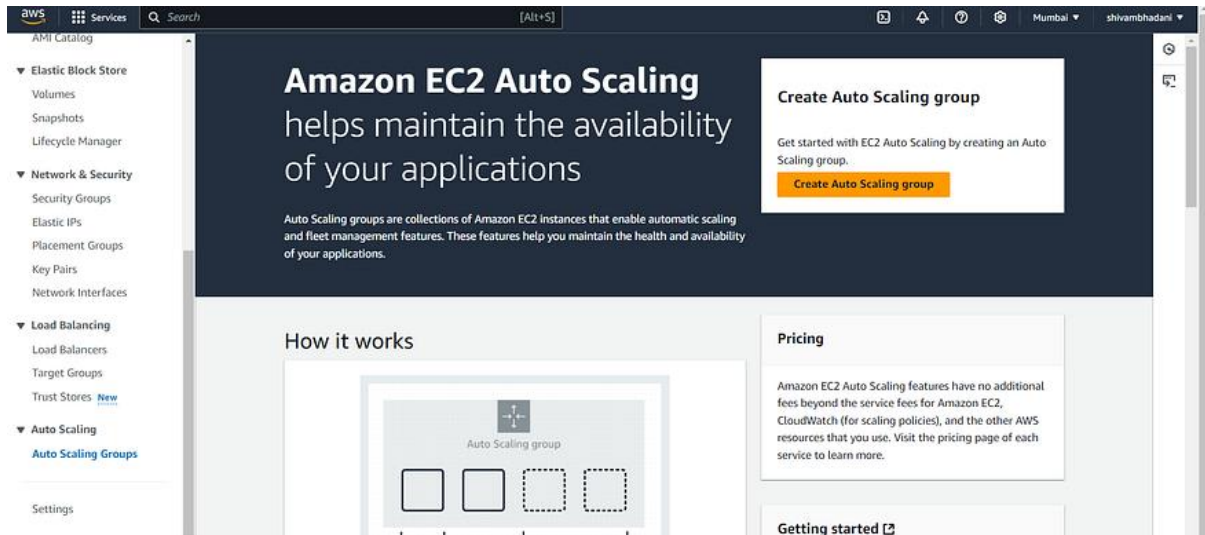


ASG

1. Go to the Auto Scaling Groups on the left sidebar and click on “Create Auto Scaling group” button.



2. Give name to this auto-scaling group (I have given shivam-ASG). Select the launch template that we have created (shivam-launch-template).

Step 3 - optional

Configure advanced options

Step 4 - optional

Configure group size and scaling

Step 5 - optional

Add notifications

Step 6 - optional

Add tags

Step 7

Review

Auto Scaling group name

Enter a name to identify the group.

shivam-ASG

Must be unique to this account in the current Region and no more than 255 characters.

Launch template

Info

ⓘ For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

shivam-launch-template

Create a launch template

Version

Default (1)

Create a launch template version

Description

My Launch Template

AMI ID

ami-067b6f633d716f14d

Key pair name

shivam-security-group

Launch template

shivam-launch-template

lt-0d9157a19bfef5c1b

Security groups

-

Security group IDs

sg-017ab878ecfda2ab0

Instance type

t2.micro

Request Spot Instances

No

Additional details

Storage (volumes)

-

Date created

Sun Oct 13 2024 19:15:31 GMT+0530 (India Standard Time)

Cancel

Next

3. Select the availability zones in which you want your new instance to be created. I selected all availability zones of ap-south-1.

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1

Choose launch template

Step 2

Choose instance launch options

Step 3 - optional

Configure advanced options

Step 4 - optional

Configure group size and scaling

Step 5 - optional

Add notifications

Step 6 - optional

Add tags

Step 7

Review

Choose instance launch options [Info](#)

Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.

Instance type requirements [Info](#)

Override launch template

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

Launch template	Version	Description
shivam-launch-template 🔗	Default	My Launch Template
Instance type		
t2.micro		

Network [Info](#)

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-0e28c412ef07e0e34

172.31.0.0/16

Default

↻

Create a VPC [🔗](#)

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

↻

ap-south-1a | subnet-02a12bd6f2dbe8f88

172.31.32.0/20

Default

✕

ap-south-1b | subnet-0b1d24493e99cbd9c

172.31.0.0/20

Default

✕

ap-south-1c | subnet-0d59498a915708c6e

172.31.16.0/20

Default

✕

Create a subnet [🔗](#)

⚠️ Your requested instance type (t2.micro) is not available in 1 Availability Zone. You may need to change the instance type or choose other Availability Zones for better resiliency. [Learn more](#) [🔗](#)

Cancel

Skip to review

Previous

Next

4. Select “Attach to an Existing Load Balancer” then choose the target group (shivam-target-group) that we have created. Whenever ASG will create a new instance, it will attach shivam-target-group to it.

[EC2](#) > [Auto Scaling groups](#) > Create Auto Scaling group

Step 1

[Choose launch template](#)

Step 2

[Choose instance launch options](#)

Step 3 - optional

Configure advanced options

Step 4 - optional

[Configure group size and scaling](#)

Step 5 - optional

[Add notifications](#)

Step 6 - optional

[Add tags](#)

Step 7

[Review](#)

Configure advanced options - *optional* [Info](#)

Integrate your Auto Scaling group with other services to distribute network traffic across multiple servers using a load balancer or to establish service-to-service communications using VPC Lattice. You can also set options that give you more control over health check replacements and monitoring.

Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

☐ No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒ Attach to an existing load balancer
Choose from your existing load balancers.

☐ Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers that you want to attach to your Auto Scaling group.

☒ Choose from your load balancer target groups
This option allows you to attach Application, Network, or Gateway Load Balancers.

☐ Choose from Classic Load Balancers

Existing load balancer target groups
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

shivam-target-group | HTTP

Application Load Balancer: shivam-alb

X

VPC Lattice integration options [Info](#)

To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.

Select VPC Lattice service to attach

☒ No VPC Lattice service
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.

☐ Attach to VPC Lattice service
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.

[Create new VPC Lattice service](#)

5. Keep everything default. We will not attach any scaling policy for now. We will do it later.

Step 4 - optional

Configure group size and scaling

Step 5 - optional

[Add notifications](#)

Step 6 - optional

[Add tags](#)

Step 7

[Review](#)

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity

Specify your group size.

1

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

Max desired capacity

1

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☒ No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☐ Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Instance maintenance policy [Info](#)

Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

Choose a replacement behavior depending on your availability requirements

Mixed behavior

☒ No policy

For rebalancing events, new instances will launch before terminating others. For all other events, instances terminate and launch at the same time.

Prioritize availability

☐ Launch before terminating

Launch new instances and wait for them to be ready before terminating others. This allows you to go above your desired capacity by a given percentage and may temporarily increase costs.

Control costs

☐ Terminate and launch

Terminate and launch instances at the same time. This allows you to go below your desired capacity by a given percentage and may temporarily reduce availability.

Flexible

☐ Custom behavior

Set custom values for the minimum and maximum amount of available capacity. This gives you greater flexibility in setting how far below and over your desired capacity EC2 Auto Scaling goes when replacing instances.

Instance scale-in protection

Scale-in protection prevents newly launched instances from being terminated by scaling activities. Make sure to remove scale-in protection for the group or individual instances when instances are ready to be terminated.

☐ Enable instance scale-in protection

Cancel

Skip to review

Previous

Next

6. Don't do anything. Click on Next, then Next, then Create Auto Scaling group button.

Step 1

[Choose launch template](#)

Step 2

[Choose instance launch options](#)

Step 3 - optional

[Configure advanced options](#)

Step 4 - optional

[Configure group size and scaling](#)

Step 5 - optional

Add notifications

Step 6 - optional

[Add tags](#)

Step 7

[Review](#)

Add notifications - optional [Info](#)

Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.

Add notification

Cancel

Skip to review

Previous

Next

Step 1

[Choose launch template](#)

Step 2

[Choose instance launch options](#)

Step 3 - optional

[Configure advanced options](#)

Step 4 - optional

[Configure group size and scaling](#)

Step 5 - optional

[Add notifications](#)

Step 6 - optional

Add tags

Step 7

[Review](#)

Add tags - optional [Info](#)

Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.

i You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group. **X**

Tags (0)

Add tag

50 remaining

Cancel

Previous

Next

Instance scale-in protection

Instance scale-in protection

☐ Enable instance protection from scale in

Step 5: Add notifications Edit

Notifications

No notifications

Step 6: Add tags Edit

Tags (0)

Key	Value	Tag new instances
No tags		

Preview code Cancel Previous Create Auto Scaling group

7. Our ASG is ready.

EC2 > Auto Scaling groups

Auto Scaling groups (1/1) Info

Launch configurations

Launch templates

Actions

Create Auto Scaling group

Q Search your Auto Scaling groups

< 1 >

☒	Name	Launch template/configuration	Instances	Status	Desired capacity	Min
☒	shivam-ASG	shivam-launch-template Version Default	1	-	1	1

Auto Scaling group: shivam-ASG

Details

Activity

Automatic scaling

Instance management

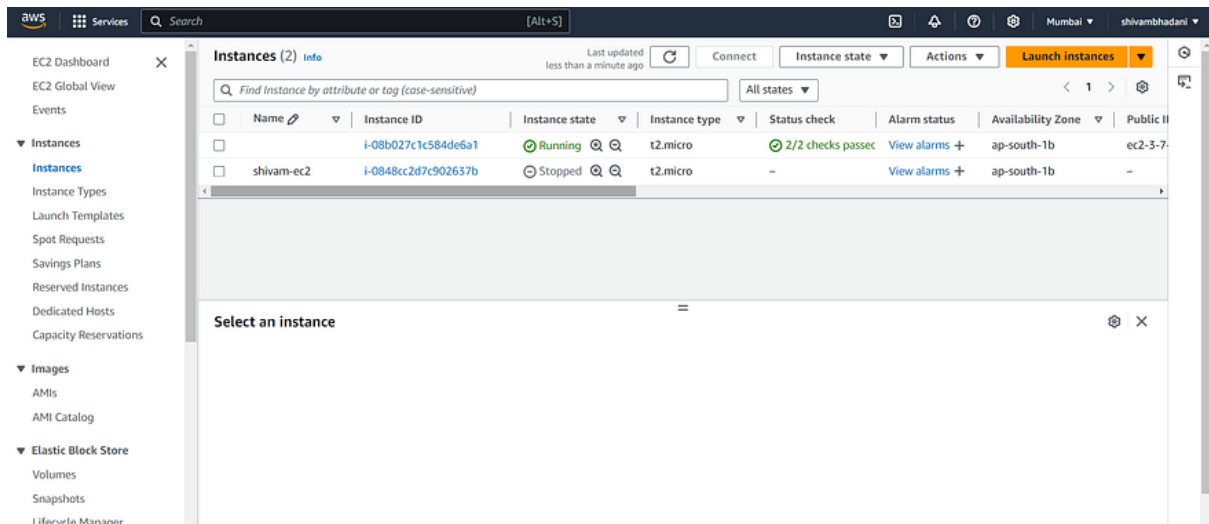
Monitoring

Instance refresh

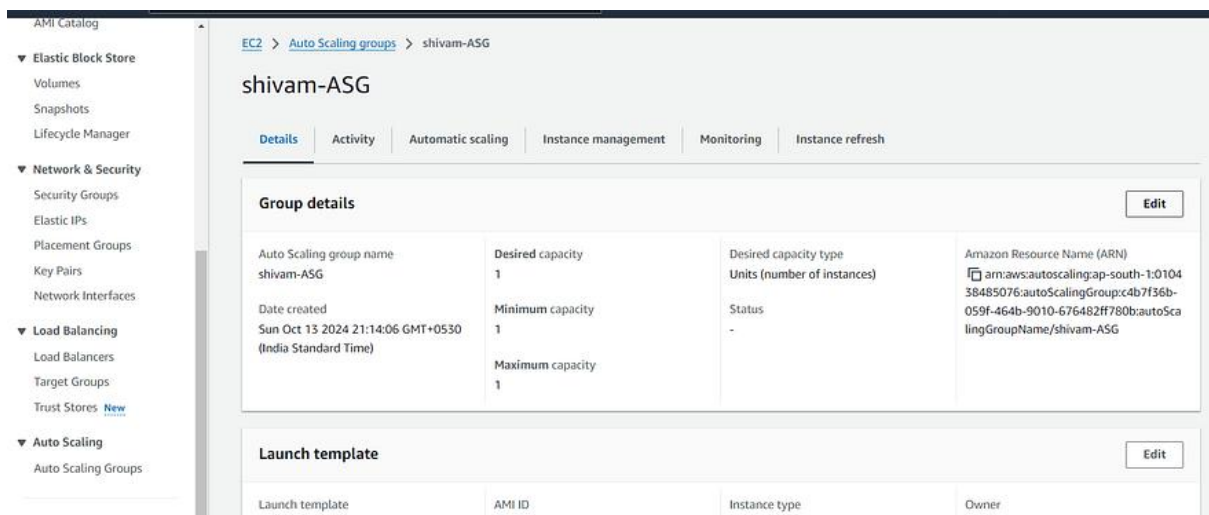
Group details Edit

Auto Scaling group name	Desired capacity	Desired capacity type	Amazon Resource Name (ARN)
shivam-ASG	1	Units (number of instances)	arn:aws:autoscaling:ap-south-1:0104

One instance is already running because we gave a minimum of 1 instance running all the time. See in step 5.



You can hit the DNS of the Load balancer on the browser and check that the site is running. 8. We will add a dynamic auto-scaling policy.



Choose your numbers. I have set the minimum no. of instances as 1 and the maximum as 5. This means if a lot of traffic comes, then it will spin up 5 instances, and when the traffic is very low, then we will only have 1 instance.

Then click on Update.

Auto Scaling groups > shivam-ASG

shivam-ASG

Details | Activity | Automatic scaling

Group size

Specify the size of the Auto Scaling group by changing the desired capacity. You can also specify minimum and maximum scaling limits.

Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances) ▼

Desired capacity
Specify your group size.

1

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity **Max desired capacity**

1 5

Equal or less than desired capacity Equal or greater than desired capacity

Cancel Update

9. Go to Automatic scaling tab then click on “create dynamic scaling policy”.

WS Services Search [Alt+S] Mumbai shivambhad

AMI Catalog

Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

Load Balancing
Load Balancers
Target Groups
Trust Stores **New**

Auto Scaling
Auto Scaling Groups

Settings

EC2 > Auto Scaling groups > shivam-ASG

shivam-ASG

Details | Activity | **Automatic scaling** | Instance management | Monitoring | Instance refresh

Scaling policies resize your Auto Scaling group to meet changes in demand. With reactive dynamic scaling policies, you can track specific CloudWatch metrics and take action when the CloudWatch alarm threshold is met. Use predictive scaling policies along with dynamic scaling policies in the following situations: when your application demand changes quickly, but with a recurring pattern, or when your EC2 instances require more time to initialize.

Dynamic scaling policies (0) Info

Create dynamic scaling policy

No dynamic scaling policies have been created

Dynamic scaling policies use real-time data to scale your group based on configurable metrics.

Create dynamic scaling policy

Predictive scaling policies (0) Info

Create predictive scaling policy

10. Give your metric. I have given that if average CPU utilization goes more than 60% then spin-up new machine. Then click on Create button.

API Catalog

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

▼ Load Balancing

Load Balancers

Target Groups

Trust Stores [New](#)

▼ Auto Scaling

Auto Scaling Groups

Settings

EC2 > Auto Scaling groups > shivam-ASG

Create dynamic scaling policy

Policy type

Target tracking scaling

Scaling policy name

Target Tracking Policy

Metric type [Info](#)

Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization

Target value

60

Instance warmup [Info](#)

300 seconds

☐ Disable scale in to create only a scale-out policy

Cancel

Create

Congratulation!! Our auto-scaling group is completed.